

What process should we use to obtain the most parsimonious set of predictor variables?

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Introduction:

Species distribution models (SDMs) can use both data collected in the field and data derived from remote sensing. This means there is often a long list of potential variables (sometimes called covariates) to include in your model. Remote sensing provides a huge amount of information that is made even more overwhelming by the large number of metrics that have become available from developed for remote sensing technologies like LiDAR. Field-work is constrained by time or resources, so decisions are often made beforehand which variables might be important for the taxa being studied. However, field data can still produce long lists of possible variables, such as ~~For example,~~ data on all co-occurring species, or data on trees across size classes, size bins ~~could create a huge list of possible variables.~~

A major challenge in species distribution modelling is deciding which variables to include in the final model or even in preliminary models. This problem is made worse by three factors. First, correlations among the hundreds of available variables are common due to ecological relationships or spatial autocorrelation. If many correlated variables are included, they add redundancy while reducing explanatory power, and may artificially inflate statistical validation using measures like 'area under the curve', often large meaning that they cannot all be included in the same candidate model. Second, many of these variables represent the same ecological process, so even if we could include them, interpreting differences in their coefficients would be challenging or arbitrary. Finally, models with too many variables will run out of degrees of freedom or fail to converge. All this means that modelling your species' distribution or abundance a global model using with the hundreds of available independent variables is not an option.

Considering this overload of information, there are two steps to selecting predictor variables for you SDM. First, you need to simply select variables that you think could represent your species habitat preferences. Second, you need to whittle down your list and select core variables for your final model. So, how do we whittle down select our predictor variables amongst this information overload? Our answer to this question takes two steps. First, how do we select the variables to include in our initial SDMs? Second, how do we select the variables to include in our final model?

How to build ~~the our~~ initial variable list?

Two recommendations exist for building our initial variable list. The **first and most often cited** is to use our ecological understanding of the species being modelled to choose reasonable variables. ~~For example, if we are building a SDM for bees with long tongues, then it makes sense to select variables that indicate how many long tubed flowers are available at each site. Variables describing the amount of grass at each site are probably less important, while variables describing the number of short tubed flowers could be marginally important. In this example you can imagine how a list of hundreds of potential variables describing a grassland plant community might be summarized into fewer than ten describing the plants most likely to be important to the long-tongued bees. Some authors (e.g. Westwood et al. 2019) develop a table of the known niche requirements of a given species, beginning with the proximate factors to fulfill a species' needs for shelter, food, and reproduction. For example, if we are building a SDM for bees with long tongues, then it makes sense to select variables that indicate~~

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If no data are available to represent these proximate factors, then, 'ultimate factors' are surmised. These are proxy variables. For example, if there is no data layer on long-tubed flowers, soil type or ecosystem classification including categories that are known to support long-tubed flower species would serve as an ultimate proxy. Of course, using this approach requires detailed biological knowledge of the species being modeled.

The **second option** is to take a quantitative approach and select one variable from each group of correlated ones. To do this, you calculate the correlation matrix of all the possible variables and look for blocks of variables that are highly correlated. If your sites are a good cross-section of the region you hope to model, you can more confidently assume that all these correlated variables are describing the same type of habitat. Once the variable groups are defined, you can either choose a variable from each to include or produce some composite variable that describes the group.

The second approach is most justifiable for generic habitat features because the link between the species habitat preference and these variables are probably indirect. For example, songbird distribution might be predicted by the basal area of a forest stand or forest age. These two variables are probably correlated, and both a proxy measure of some unknown or unmeasured variable(s) that the birds need. Including only one of them in the initial model would be justified in this case but would not be justified if forest age and basal area were uncorrelated.

How to select the variables in our final model?

We can use model selection methods to determine which variables we keep in our final model. These methods begin with assessing the fit of our global model using residual plots to determine whether the model is a good fit. You shouldn't see any patterns in the residuals. Sometimes you might have too many variables and still cannot build a **global** model. In this case, you should explore the fit of several sub-global models. Make sure to examine the residuals across space and check for overdispersion. If the model fit is poor, you will need to go back to your variable list to look for excluded variables or consider a different modelling approach. If the model fit statistics look OK, then you can proceed with model selection.

Information criteria is often used as a model selection tool for SDMs, although removing variables with **insignificant** is also OK. If you are using information criteria, make sure to correct them for overdispersion and keep an eye out for uninformative variables. The latter are variables with small coefficient estimates and large standard errors that sometimes end up in models with lower information criteria scores. You can use backwards selection from a global model or run all possible model subsets to find the final model with the lowest information criteria score. Your final model could be the model with the lowest information criteria score or a model average combination of the best **models**.

Using more than one approach is key**Summary:**

Commented [1]: I think 'global model' might need defining on first use, just so folks aren't confusing it with like, global in terms of geographic coverage

Commented [2R1]: Agreed. If I'm thinking along the right lines, I would add something like: The global model is the base model that was created and trained using field-sampled data of which predictions can be transferred to an area with little to no reference data.

Commented [3]: Word missing here I think

Commented [4]: Is there a paper you can link to that uses this approach?

Species distribution models are challenging because we often have an overwhelming number of variables from remote sensing or field surveys to include in our models. Selecting the most parsimonious list of variables is not so complex once we have a well-specified global model. ~~The challenge is how to~~ get from the list of hundreds of possible variables to a list of tens of variables to include in ~~our~~ global model. ~~we suggest combining. Our suggestion is to use a mixture of~~ your understanding of the species ecology to focus on the most likely variable ~~set to make an initial variable list. Then, s and~~ examine the correlation between the initial variable list to identify which ones are providing new information about the habitat at your study sites.